Research in Developmental Disabilities 172 (2026) 105276

Contents lists available at ScienceDirect

Research in Developmental Disabilities

journal homepage: www.elsevier.com/locate/redevdis

Effects of exercise on social dysfunction in children and adolescents with ADHD: A systematic review and network, dose-response meta-analyses☆

Xiang Gao [a,1], Meng Zhang [a,1], Changshuang He [a], Zongyu Fan [a], Keyu Pan [a], Huan Huang [b], Shanru Jin [a], Yiting Wei [a], Wanqun Xu [a], Minghui Quan [a,*]

[a] School of Exercise and Health, Shanghai University of Sport, Shanghai 200438, China
[b] Department of Rehabilitation Sciences, Faculty of Education, East China Normal University, Shanghai 200062, China

ARTICLE INFO

Keywords:
Attention-deficit/hyperactivity disorder
Social functioning
Children
Network meta-analysis
Dose-response relationship

ABSTRACT

Background: Children with ADHD frequently face social dysfunction hindering neurodevelopment. While exercise shows promise, the relative efficacy of specific modalities and precise dose-response relationships remain undefined. This study compares exercise modalities to identify the optimal dose for enhancing social functioning.

Methods: Seven databases were systematically searched from inception to March 2025 for randomized controlled trials of exercise interventions targeting social outcomes in children with ADHD. Exercise interventions were categorized by modality (closed-skill exercise, open-skill exercise, multicomponent exercise and mind-body therapy). Doses were harmonized to MET-min/week. Bayesian network meta-analysis (Hedges' g) synthesized relative efficacy, while a 4-knot natural-spline model identified the optimal dose-response range.

Results: Thirteen trials including 703 participants aged 6–13 years were eligible. Closed-skill exercise produced the largest improvements in social functioning (Hedges' g = −0.96, 95% CrI: −1.18 to −0.75). Multicomponent exercise (Hedges' g = −0.74, 95% CrI: −1.13 to −0.34) and mind–body therapy (Hedges' g = −0.53, 95% CrI: −0.79 to −0.28) showed moderate, clinically meaningful effects, whereas open-skill exercise had small effects (Hedges' g = −0.16, 95% CrI: −0.36 to −0.04). Dose-response analysis indicated an inverted U-shaped relationship, with a minimum effective dose of 660 MET-min/week and an optimal dose of 1290 MET-min/week. Both age and gender had a significant impact on the outcomes.

Abbreviations: ADHD, Attention-Deficit/Hyperactivity Disorder; CINeMA, Confidence in Network Meta-Analysis; CrI, Credible Interval; DSM, Diagnostic and Statistical Manual of Mental Disorders; GRADE, Grading of Recommendations Assessment, Development and Evaluation; ICD, International Classification of Diseases; MCID, Minimal Clinically Important Difference; MeSH, Medical Subject Headings; MET, Metabolic Equivalent of Task; NMA, Network Meta-Analysis; PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses; RCT, Randomized Controlled Trial; ROB, Risk of Bias; RoBMA, Robust Bayesian Meta-analysis; SUCRA, Surface Under the Cumulative Ranking Curve; WHO, World Health Organization.

☆ PROSPERO registration number: (CRD42025632382).

* Correspondence to: School of Exercise and Health, Shanghai University of Sport, 399 Chang Hai Road, Yang Pu district, Shanghai 200438, China.

E-mail address: quanminghui@sus.edu.cn (M. Quan).

[1] These two authors have contributed equally to this work.

https://doi.org/10.1016/j.ridd.2026.105276
Received 13 December 2025; Received in revised form 26 March 2026; Accepted 27 March 2026
Available online 2 April 2026
0891-4222/© 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Conclusion: Closed-skill and multicomponent exercise appear most effective for improving social functioning in children with ADHD, with benefits maximised at around 1290 MET-min/week.

1. Introduction

Attention Deficit/Hyperactivity Disorder (ADHD) is the most prevalent neurodevelopmental disorder in childhood, affecting approximately 5–7% of school-age children and adolescents worldwide (Salari et al., 2023). Its core symptoms—inattention, impulsivity, and hyperactivity—adversely affect academic performance, psychological development, and social integration. While these symptoms typically emerge in early childhood, research indicates that over 50% of affected individuals continue to experience functional impairments into adulthood (Danielson et al., 2024), often manifested through persistent social dysfunction and diminished quality of life (Samosh et al., 2024). Consequently, the impaired social functioning observed among children and adolescents diagnosed with ADHD, particularly evident in peer interactions, represents a significant clinical concern requiring attention (Faraone et al., 2021).

Social dysfunction in ADHD is a multifaceted construct that manifests across three primary domains: (1) social skills performance deficits (the 'knowing-doing' gap), (2) impaired social information processing, and (3) diminished social reciprocity and peer relationship stability (Bunford et al., 2015; McQuade & Hoza, 2015). Approximately 60% of children and adolescents with ADHD experience significant social impairment, which not only hinders immediate development but also elevates the risk of comorbid internalizing disorders, such as anxiety and depression, in later life (Nijmeijer et al., 2008). These challenges are further compounded by comorbid conditions like Oppositional Defiant Disorder (ODD) or Conduct Disorder (CD), which exacerbate symptom severity and psychosocial impairment (Eskander, 2020), engendering a pernicious cycle that negatively impacts familial harmony and future vocational prospects.

At present, pharmacological treatment remains the primary approach for managing ADHD, as evidenced by the utilisation of central nervous system stimulants, including extended-release methylphenidate formulations (Biederman & Faraone, 2005). Non-pharmacological interventions encompass health education, cognitive behavioural therapy (Safren et al., 2005), alongside comprehensive approaches including lifestyle training, mindfulness and meditation, and occupational therapy. Although medication has been demonstrated to be efficacious in the alleviation of core ADHD symptoms (such as attention deficits and impulse control), there is currently no evidence to suggest that it can enhance positive social behaviours or normalize peer status (Hoza et al., 2005). Moreover, parents frequently express concerns regarding its side effects, including reduced appetite and sleep disturbances (Nafees et al., 2014; Stein et al., 2012). Conversely, the focus on exercise intervention as a low-risk, highly feasible, and multifaceted non-pharmacological approach is increasing (Cerrillo-Urbina et al., 2015). Research indicates that exercise not only effectively improves core symptoms but also positively impacts social functioning by providing structured social settings that enhance teamwork abilities and social skills in children and adolescents with ADHD (Pontifex et al., 2013; Zhu et al., 2023). Consequently, exercise is regarded as a potentially effective complementary treatment for ADHD.

Despite the exploration of the impact of exercise on the social functioning of children and adolescents with ADHD in existing research, a systematic and quantitative synthesis is warranted for several reasons. First, a substantial body of new evidence has been published since the lats meta-analysis conducted in 2015, requiring an urgent update of the current knowledge base. Second, previous reviews often failed to evaluate social functioning as an independent primary outcome, typically subsuming it under broader "behavioral improvement" without dedicated indicator analyses (Cornelius et al., 2017). Moreover, there remains a paucity of systematic comparisons regarding the relative efficacy of different exercise modalities and a lack of evidence concerning the dose-response relationship between intervention parameters (e.g., frequency, intensity, total volume) and social outcomes. To address these gaps, the present study employs network meta-analysis and dose-response modeling to systematically evaluate the efficacy of various exercise types and identify optimal intervention dosages, thereby providing a theoretical foundation and practical guidance for clinical rehabilitation.

2. Method

2.1. Protocol registration

This review strictly adheres to the Preferred Reporting Items for Systematic Reviews and Network Meta-Analyses (PRISMA-NMA) guidelines (Page et al., 2021) and the Cochrane Collaboration manual (Jpt, 2008).The study protocol has been registered with the PROSPERO International Prospective Systematic Reviews Register (Registration number: CRD42025632382). As this study involves only a review and synthesis of published literature without human or animal experimentation, approval from an institutional review board is not required.

2.2. Information and search strategy

A comprehensive search strategy was employed, encompassing eight prominent electronic databases, namely MEDLINE, Embase, the Cochrane Library, Web of Science, Scopus, SPORT Discus, China National Knowledge Infrastructure, and CINAHL. The search was

conducted from inception to March 2025, with no restrictions applied to the publication date or language. The search strategy was formulated through the combination of keywords and Medical Subject Headings terms, including 'children', 'adolescents', 'attention-deficit/hyperactivity disorder', 'ADHD', 'physical activity' and 'exercise'. The detailed search strategy is presented in Supplement 1. In addition to conducting database searches, a cross-check was performed on reference lists of key reviews, and Google Scholar was searched using the same search terms to identify relevant citations within included studies.

2.3. Inclusion and exclusion criteria

Studies were included if they met all the following criteria:

1) Participants: Children or adolescents aged 6–18 years with ADHD (diagnosed according to DSM (Diagnostic and statistical manual of mental disorders: DSM-5™, 5th ed, 2013) or ICD (Organization, 2004) criteria);
2) Duration of exercise intervention: $\geq$ 4 weeks (Jonsson et al., 2019) ;
3) Outcomes: It is imperative that assessments of social functioning impairments in children and adolescents be included;
4) Study design: The present study exclusively considers randomized controlled trials (RCTs). Research is required that compares different types or intensities of exercise, or that compares exercise groups with control groups.
5) The publication has been disseminated through the medium of peer-reviewed academic journals, with the written content being presented in either English or Chinese.

The following criteria are to be considered as exclusionary:

1) Studies using healthy children as control groups ;
2) Interventions based on multimodal approaches, such as exercise combined with psychotherapy or behavioural interventions ;
3) Identification of duplicate publications ;
4) Studies with erroneous, missing, or unobtainable data.

2.4. Study selection and data extraction

We imported the retrieved studies into Endnote X9 and removed duplicate trials. Two reviewers (X.G. and M.Z.) independently screened study titles, abstracts, and full texts, extracting data on study characteristics, participant characteristics, interventions, and study outcomes. For studies reporting two or more measurement time points, we selected only baseline and post-intervention measurements. Where data were presented graphically, we attempted extraction using GetData Graph Digitizer. For studies where data could not be obtained, we will make three attempts to contact the corresponding author within six weeks to request the data. In the event of disagreement, we will resolve the issue through discussion or by consulting a third independent author (M.H.Q.) until consensus is reached.

2.5. Data coding and management

Exercise interventions were categorized into four distinct modalities: closed-skill exercise, open-skill exercise, multicomponent exercise, and mind-body therapy. The classification was based on the predictable nature of the environment and the specific motor demands of each activity. Detailed definitions and the classification criteria for each included study are provided in Supplement 2. The exercise intensity unit is defined as the product of the specific exercise intensity (task metabolic equivalent [MET]), weekly frequency, and single-session duration based on primary intervention time (excluding warm-up and cool-down), expressed as MET-min/week (Li et al., 2025). Duration used in the study was not explicitly defined; mean total duration was calculated using study averages and progressive protocols. MET values were assigned according to A Youth Compendium of Physical Activities (Butte et al., 2018); for unlisted activities, the closest equivalent activity in the guidelines was used (Li et al., 2025). For multi-component interventions, overall intensity was calculated as the duration-weighted average MET value of each component. Exercise intensity ategories were predefined as follows: light (<3.0 MET), moderate (3.0–6.0 MET), and vigorous (>6.0 MET).

2.6. Quality assessment

Risk of bias assessment employed the Cochrane Risk of Bias tool for randomized controlled trials (RoB) (Higgins et al., 2011), with feasibility evaluation conducted using the Confidence IN Network Meta-Analysis (CINeMA) tool (Nikolakopoulou et al., 2020). CINeMA is a tool designed to assess the confidence in the results of network meta-analyses (NMAs). It assists researchers in evaluating NMA outcomes and determining their reliability. More specifically, CINeMA employs a series of criteria to assess consistency between direct and indirect evidence, alongside other potential sources of bias, thereby providing an overall confidence assessment for NMA results. The assessment encompasses selection bias, performance bias, attrition bias, detection bias, and reporting bias. Following Cochrane Handbook criteria (Higgins et al., 2011), each bias domain is judged as “yes”, “no”, or “unclear” to indicate whether sufficient measures were taken to prevent each potential source of bias within each study. Two independent reviewers conducted the risk of bias and credibility assessments independently. Any discrepancies were resolved through discussion or by arbitration by a third author.

2.7. Summary measures and synthesis

To measure the magnitude of effects across different interventions, this study employs Hedges'g as a standardized effect size indicator, interpreted as follows: small effect (~0.2), moderate effect (~0.5), large effect (~0.8) (Cohen, 1988). Hedges'g effectively corrects for bias arising from small sample sizes and facilitates comparisons across studies with varying sample sizes. Effect sizes were calculated based on the standardized mean difference (SMD) and standard deviation (SD) from each study, ensuring a consistent measure of intervention effects. For studies lacking SD data, we estimated the standard deviation using standard error, 95% confidence intervals, p-values, or t-statistics. Following the Cochrane Collaboration's recommended bias risk assessment tool, we employed a correlation coefficient of 0.5 to calculate SD differences (Higgins et al., 2019). Furthermore, referencing prior research, we defined 0.5 SD units as the minimum clinically important difference (MCID) (Norman et al., 2003).

The primary analysis employed a Bayesian multilevel network meta-analysis model based on arms, assuming each observed effect size represents a sample from a broader true effect distribution. To account for sampling error and heterogeneity in effect sizes across studies, we utilized a random-effects model to estimate the effect of exercise interventions, thereby enhancing the stability and

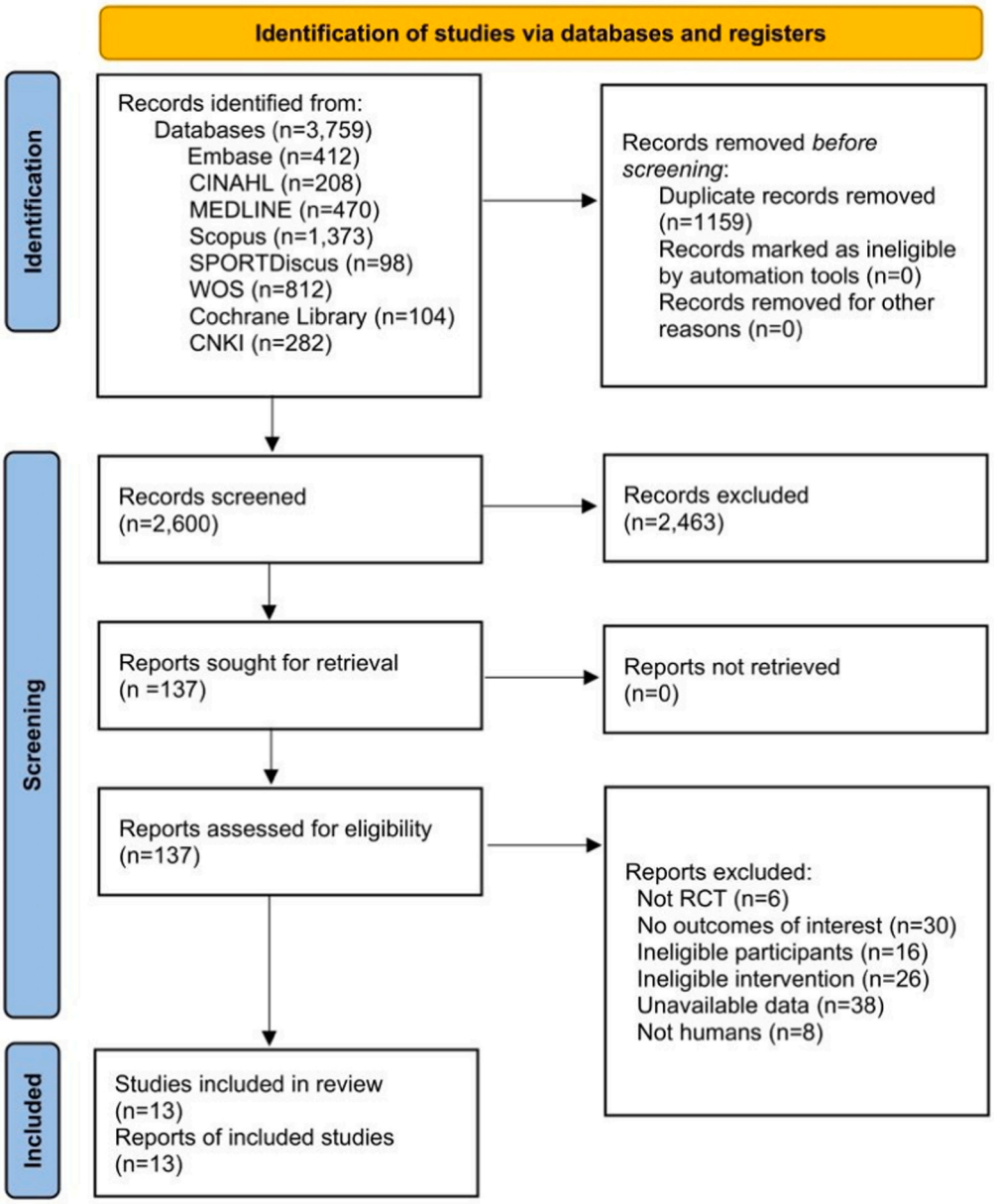


Fig. 1. PRISMA flow diagram of study selection. CNKI China National Knowledge Infrastructure; RCT Randomized controlled trial; WOS, Web of Science.

reliability of the analytical results. The Bayesian analysis fitted a multilevel random-effects model using the R package 'brms' (Bürkner, 2017), with model intercepts specified using weak prior information (population effect size $\mu \sim [0, 1]$, between-studies heterogeneity $\tau \sim [0, 1]$). Model convergence was assessed using the potential scale reduction factor (PSRF) (Gelman et al., 1995), with PSRF values below 1.05 indicating good model fit. To address dependency among multiple effect sizes within the same study, we employed a multilevel model to account for intra-study dependence of multiple effect sizes.

The ranking of different interventions was assessed by calculating the Sum of Areas Under the Cumulative Rank Curve (SUCRA) values (Salanti et al., 2011). The similarity in the distribution of key effect modifiers between comparison groups was also examined in order to validate the transferability assumptions of network meta-analysis. Furthermore, the consistency between direct and indirect comparisons was evaluated using the node partitioning method within the 'netmeta' software package (Rücker & Schwarzer, 2014), and network consistency was tested using the 'by treatment design' model. All statistical analyses were conducted within the R environment (version 4.4.1).

2.8. Additional analyses

A random-effects model was employed, with the brms package (Bürkner, 2017) serving as the underlying framework for conducting a non-linear dose-response analysis (natural spline, 4 knots). The present analysis modelled the relationship between weekly exercise dose and the duration of the exercise intervention, and social dysfunction in children and adolescents diagnosed with ADHD.

Table 1
Characteristics of included studies.

Studies	Country	Funded	Participant samples, Female (%)	Age (mean ± SD) (years)	Intervention	Intensity prescribed (MET)	Exercise dose (MET/week)	Length (weeks)
Ahmed & Mohamed, 2011	Egypt	N	EG: 42, 35.71 CG: 42, 35.71	EG: 13.90 ± 1.60 CG: 13.80 ± 1.70	EG: Closed-skill exercises CG: no treatment	EG: 5 CG: 0	EG: 525 CG: 0	10
Bustamante et al., 2016	USA	Y	EG: 19, 32.00 CG: 16, 31.00	EG: 9.40 ± 2.20 CG: 8.70 ± 2.00	EG: Multicomponent exercises CG: Sedentary activity	EG: 6 CG: 1	EG: 1800 CG: 300	10
Yumin & Liang, 2016	China	N	EG: 18, 55.56 CG: 18, 50.00	EG: 8.48 ± 0.52 CG: 8.51 ± 0.50	EG: Mind-body therapy CG: No treatment	EG: 3 CG: 0	EG: 594 CG: 0	16
García-Gómez et al., 2016	Spain	Y	EG: 9, 33.33 CG: 5, 33.33	EG: 10.65 ± 1.50 CG: 10.20 ± 2.38	EG: Open-skill exercises CG: No treatment	EG: 6 CG: 0	EG: 495 CG: 0	12
Hattabi et al., 2022	Tunisia	N	EG: 20, 15.00 CG: 20, 10.00	EG: 9.95 ± 1.31 CG: 9.75 ± 1.33	EG: Closed-skill exercises CG: No treatment	EG: 7 CG: 0	EG: 1512 CG: 0	12
Hoza et al., 2015	USA	Y	EG: 49, 44.20 CG: 45, 49.00	EG: 6.83 ± 0.96 CG: 6.83 ± 0.96	EG: Open-skill exercises CG: Sedentary activity	EG: 3 CG: 1	EG: 420 CG: 155	12
Jensen & Kenny, 2004	Australia	N	EG: 11, 0.00 CG: 8, 0.00	EG: 10.63 ± 1.78 CG: 9.35 ± 1.70	EG: Mind-body therapy CG: Education	EG: 3 CG: 1	EG: 180 CG: 15	20
Kang et al., 2011	Korea	Y	EG: 16, 0.00 CG: 16, 0.00	EG: 8.40 ± 0.90 CG: 8.60 ± 1.20	EG: Multicomponent exercises CG: Education	EG: 6 CG: 1	EG: 713 CG: NA	12
Oh et al., 2018	Korea	Y	EG: 17, 11.76 CG: 17, 5.88	EG: 8.30 ± 1.48 CG: 8.00 ± 1.70	EG: Open-skill exercises CG: Medication	EG: 4 CG: 0	EG: 456 CG: 0	12
Pan et al., 2016	China	Y	EG: 16, 0.00 CG: 16, 0.00	EG: 8.93 ± 1.49 CG: 8.87 ± 1.56	EG: Open-skill exercises CG: No treatment	EG: 4 CG: 0	EG: 588 CG: 0	12
Sabzi et al., 2021	Iran	N	EG: 20, 0.00 CG: 20, 0.00	EG: 9.60 ± 0.50 CG: 9.50 ± 0.30	EG: Closed-skill exercises CG: No treatment	EG: 5 CG: 0	EG: 450 CG: 0	8
Silva et al., 2020	Brazil	Y	EG: 18, 20.00 CG: 15, 40.00	EG: 12.00 ± 2.00 CG: 12.00 ± 1.00	EG: Closed-skill exercises CG: No treatment	EG: 4 CG: 0	EG: 606 CG: 0	8
Wang, 2023	China	NA	EG: 15, NA CG: 15, NA	EG: 7.01 ± 0.84 CG: 7.01 ± 0.84	EG: Mind-body therapy CG: No treatment	EG: 3 CG: 0	EG: 180 CG: 0	20

Notes: Y, yes; N, no; NA, not available; CG, control group; EG, exercise group; SFD, social function disorder; USA, the United States.

In addition, participant characteristics (age, gender) and funding availability were considered as potential moderators, with the aim of assessing the robustness of the model. Furthermore, a multilevel Egger's regression test (Rodgers & Pustejovsky, 2020) was conducted in order to examine potential small-study effects. In addition, to incorporate a more state-of-the-art publication-bias-adjusted approach, a supplementary robust Bayesian meta-analysis (RoBMA) was performed (Bartoš et al., 2022). RoBMA model-averages across models assuming the presence versus absence of an overall effect, heterogeneity, and publication bias, thereby yielding publication-bias-adjusted pooled estimates. Because some studies reported multiple SFD symptom indicators/tasks, task-specific effect sizes within the same treatment arm were first aggregated at the study-arm level to avoid double-counting. The primary aggregation assumed a within-study correlation of $\rho = 0.6$, with additional sensitivity analyses using $\rho = 0.3$ and $\rho = 0.8$. The resulting study-level pairwise effect sizes and their standard errors were then entered into the RoBMA.

3. Results

3.1. Study selection and characteristics

The search process yielded a total of 3759 citations, of which 2600 were retained subsequent to the removal of duplicates (see Fig. 1). Following the rigorous screening process of 2600 titles and abstracts, as well as 137 full texts, 13 randomized controlled trials were identified as meeting the inclusion criteria for this systematic review (Supplement 14). The studies encompassed a total of 703 participants with an average age ranging from 6.83 to 13.85 years, with female participants accounting for only approximately 2.6% of the study population. Six of the included studies originated from high-income countries: two from the United States, two from South Korea, one from Australia, and one from Spain. Furthermore, two studies were conducted in China, one in Egypt, one in Brazil, and one in Iran. The characteristics of the included studies and the features of the participant and treatment groups are detailed in Table 1 and Supplement 3.

3.2. Study bias and quality for evidence

Supplement 4 presents the reporting bias tables, risk of reporting bias, and bias items for each study. The studies were then categorized into three risk levels: low risk of bias (0.00%), some risk of bias (84.62%), and high risk of bias (15.38%). The results of the CINeMA study, as outlined in Supplement 12, indicate that the quality of the evidence supporting intervention measures is categorised into three levels: low (40%), moderate (40%), and high (20%). The primary reasons for downgrading the quality ratings are as follows: risk of bias, the small number of direct comparison studies (<10), high heterogeneity, and minimal overlap in CRLs.

3.3. Comparative efficacy of different exercise modalities

Fig. 2 presents comparative results across different exercise intervention types. Compared with the control group, the Closed-skill exercises group exhibited the highest number of participants and direct comparisons. The Closed-skill exercises group (Hedges'g = −0.96, 95% CrI: −1.18 to −0.75, k = 4, SCURA = 0.54) demonstrated a clinically significant effect in improving social dysfunction among children and adolescents with ADHD. The Multicomponent exercises group (Hedges'g = −0.74, 95% CrI: −1.13 to −0.34, k = 2, SCURA = 0.63) and Mind-body therapy (Hedges'g = −0.53, 95% CrI: −0.79 to −0.28, k = 3, SCURA=0.80) yielded moderate-effect therapies, likewise achieving clinically significant effects. The open-skill exercises group (Hedges'g = −0.16, 95% CrI: −0.36 to −0.04, k = 4, SCURA = 0.38) exhibited a smaller effect size. I^2 was 67.5% (moderate to high), with an overall Q-statistic for

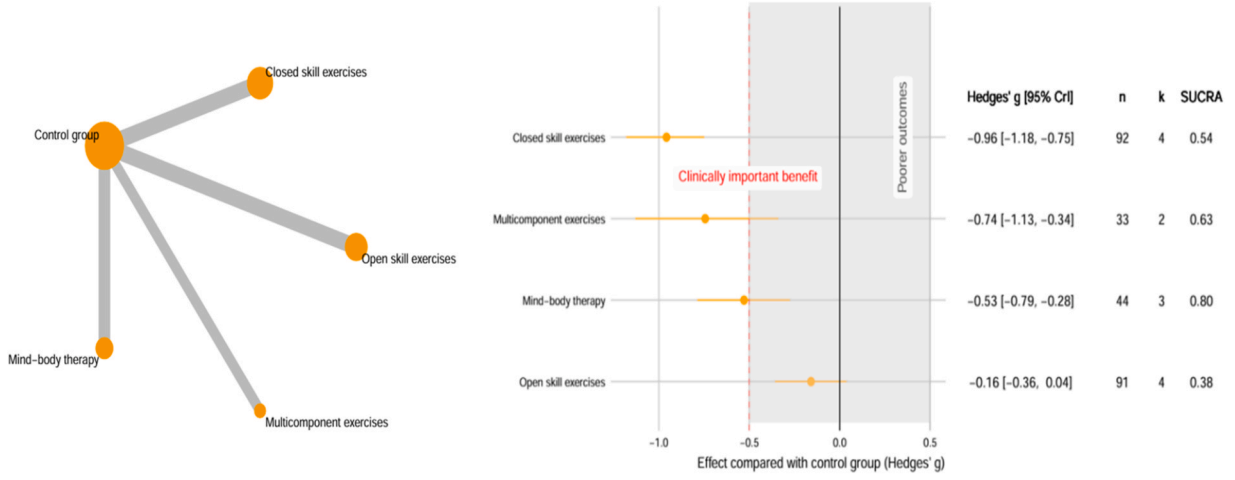


Fig. 2. Network diagrams depicting the direct and indirect comparisons for the social function disorder symptom network meta-analyses and corresponding Bayesian ranking panel plots. SUCRA, surface under the cumulative ranking curve; CrI, credible interval.

heterogeneity of 336.61 ($P < 0.001$). Conventional assessments of small-study effects at the network level did not indicate clear evidence of funnel plot asymmetry, and the multilevel Egger's regression test was not statistically significant ($F_{1,80} = 1.5783$, $p = 0.2127$) (Supplement 11). To further examine the robustness of the overall findings, a supplementary publication-bias-adjusted sensitivity analysis was conducted using RoBMA based on study-level pairwise effect sizes aggregated across task-specific outcomes.

Under the primary within-study correlation assumption ($\rho = 0.6$), RoBMA yielded moderate evidence for publication bias (inclusion BF = 5.64) and strong evidence for heterogeneity (inclusion BF = 18.44), whereas evidence for a non-zero pooled effect was inconclusive (inclusion BF = 0.86). After adjusting for potential bias, the estimated pooled effect size was 0.146 (95% CrI: −0.509,0.820). Sensitivity analyses using alternative within-study correlations ($\rho = 0.3$ and $\rho = 0.8$) yielded a broadly similar overall pattern, with evidence for publication bias remaining present and the credible interval of the adjusted pooled effect continuing to include zero. While the primary NMA suggested a significant effect, the bias-adjusted estimate from the RoBMA framework crossed the null value, indicating that the observed significance might be sensitive to potential publication bias (Supplement 13).

3.4. The dose-response relationship and clinical MCID thresholds

Fig. 3 illustrates the non-linear relationship between weekly exercise dose and therapeutic efficacy for social dysfunction, revealing a U-shaped dose-response relationship characterized by an inverted U-curve. The MCID dose was determined to be 660 MET-min/week

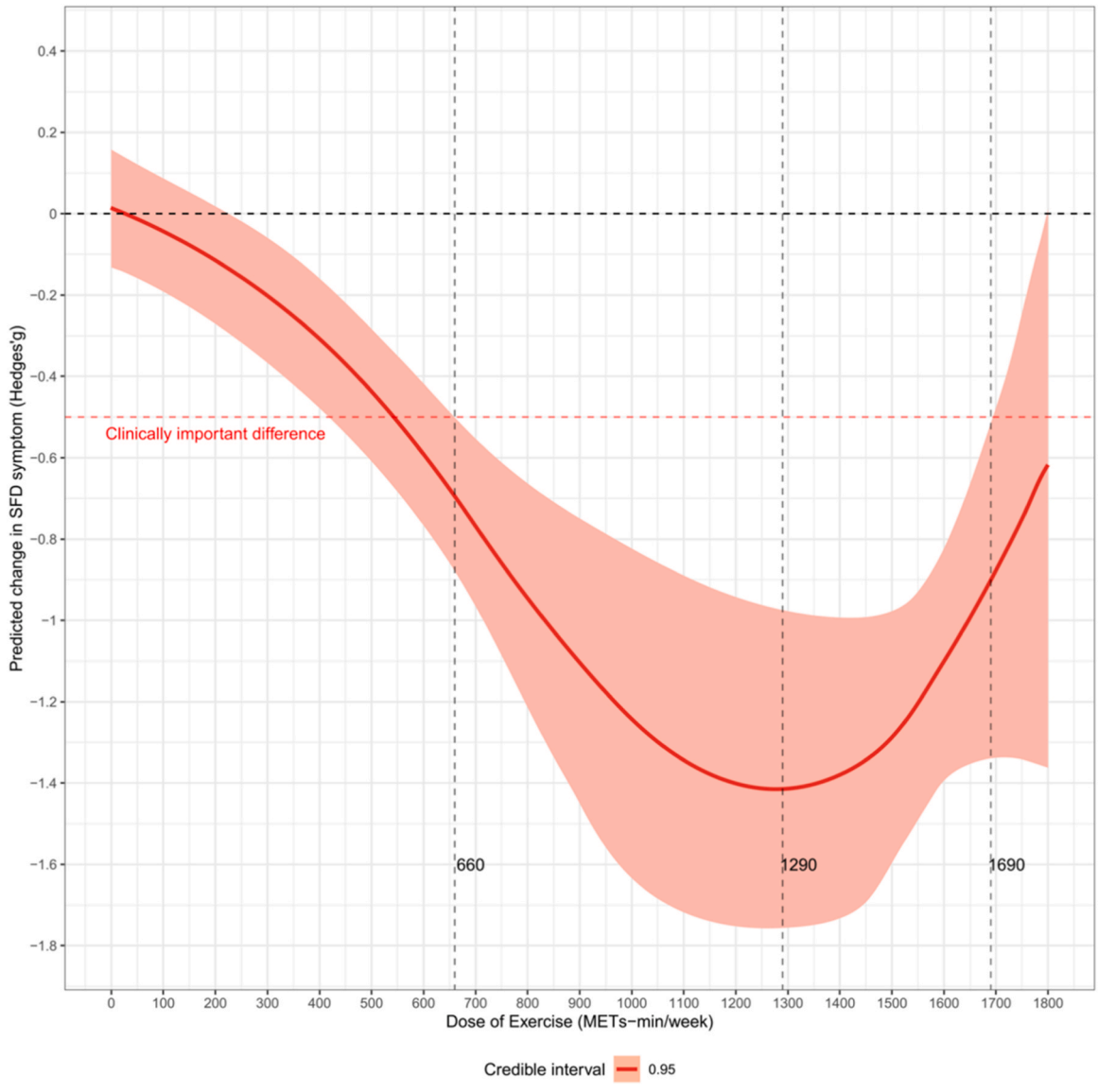


Fig. 3. Dose-response relationship between exercise and improvement in social function disorder symptom in children and adolescents with ADHD. SFD, social function disorder.

(Hedges'g = −0.7, 95% CrI = −0.51 to −0.89), equivalent to 16–31 min of moderate-intensity exercise daily. The optimal dose was 1290 MET-min/week (Hedges'g = −1.43, 95% CrI = −0.98 to −1.89), corresponding to 31–61 min of moderate-intensity exercise daily.

A detailed investigation into the duration of intervention and its efficacy in treating social dysfunction was conducted, revealing a non-linear relationship that exhibited a positive correlation trend. This relationship demonstrated that as intervention duration increased, therapeutic effectiveness diminished (see Supplement 8). Additionally, the broader 95% credible interval (CrI) in the subsequent phases signifies an elevated degree of uncertainty with protracted intervention. It is recommended that future research incorporate additional studies with a view to further validating and refining these trends. This would enhance the accuracy and reliability of predictions.

3.5. Moderation by participants and study characteristics

Analysis of the efficacy of exercise on social functional impairment revealed that compared to models considering activity type alone ($R^2 = 0.273$), both age ($R^2 = 0.374$) and gender ($R^2 = 0.334$) models increased explanatory power, whereas the funded model ($R^2 = 0.261$) showed reduced explanatory power (Supplement 9). Women may experience greater improvements in social functional impairment within the Mind-body therapy group. Conversely, men may achieve greater improvements in social functional impairment within the Multicomponent exercises, Closed-skill exercises, and Open-skill exercises groups (Fig. 4). The relationship between exercise intervention effects and age also varied. Treatment efficacy diminished with increasing age in the Multicomponent exercises, Closed-skill exercises, and Mind-body therapy groups, whereas it remained relatively stable in the Open-skill exercises group (Fig. 4).

4. Discussion

4.1. Summary of principal findings

The present study indicates that exercise interventions can improve social functioning in children and adolescents with ADHD, with interventions characterized by higher levels of structuralization demonstrating more pronounced effects. Specifically, network meta-analysis reveals that closed-skill exercises possess the most potent efficacy in enhancing social functioning, followed by multicomponent exercise and mind-body therapy, while open-skill exercise exhibit comparatively diminished effects. Moreover, this research

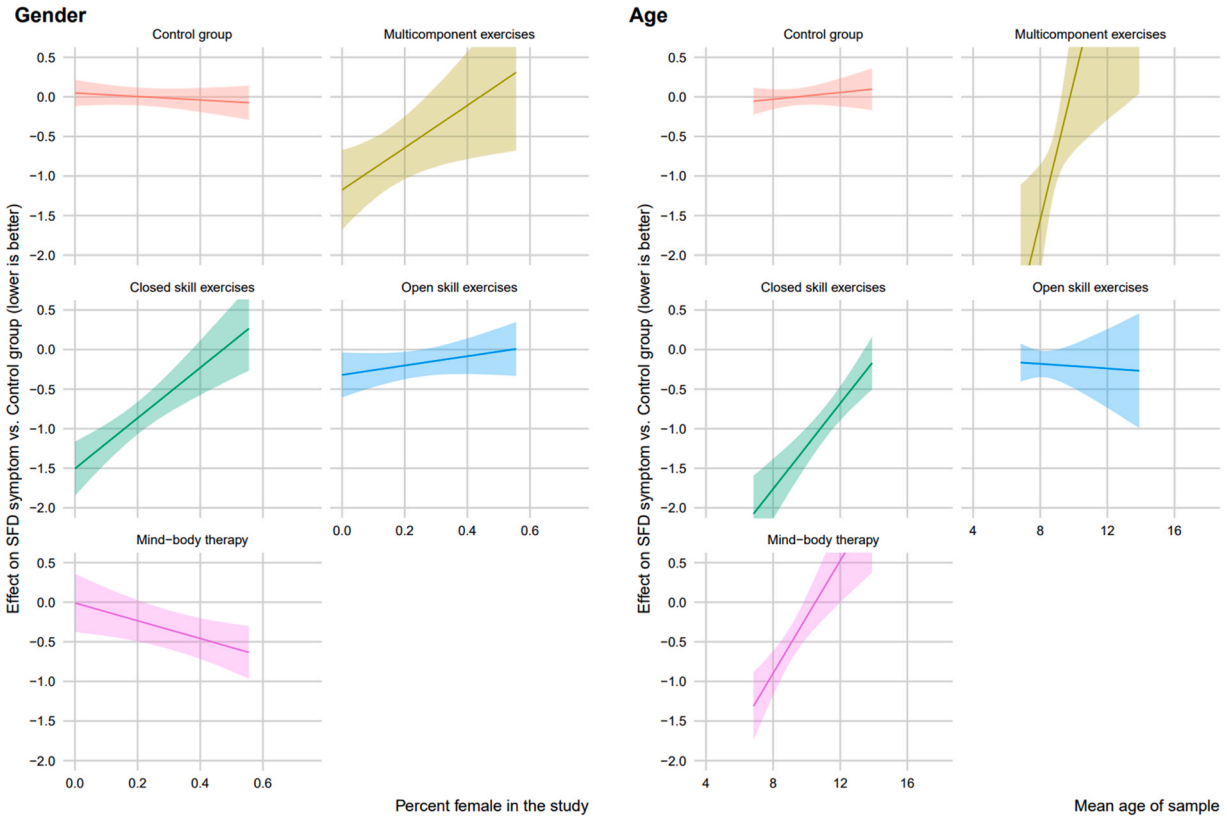


Fig. 4. Moderation by gender and age. SFD, social function disorder.

identifies a non-linear, U-shaped dose-response relationship between exercise volume and social function improvement. We established a clinically significant threshold for the Minimal Clinically Important Difference (MCID) at 660 MET-min/week (equivalent to 16–31 min of daily moderate-intensity exercise) and identified an optimal therapeutic dose at 1290 MET-min/week (equivalent to 31–61 min of daily moderate-intensity exercise). Extrapolated analysis further revealed that gender and age serve as critical moderating factors, suggesting that individual phenotypic characteristics significantly influence treatment responsiveness. Taken together, these findings provide quantitative support for the potential clinical application of exercise interventions in addressing social dysfunction in pediatric ADHD populations.

4.2. Theoretical mechanisms of social impairment

The social dysfunction observed in children and adolescents with ADHD is not an isolated phenomenon but is intricately intertwined with core symptomatology and underlying executive function deficits (Nijmeijer et al., 2008). Research has demonstrated that deficits in inhibitory control, cognitive flexibility, and working memory make it more challenging for these children and adolescents to regulate their emotions, adhere to social norms, and express themselves appropriately in real-world interactions, thereby significantly limiting their social capabilities (Christiansen et al., 2019; Zhu et al., 2023). While these neuropsychological mechanisms offer substantive explanations for the etiology of social impairment, the enhancement of social functioning is a multidimensional trajectory that extends beyond the mere mitigation of core symptoms. Nevertheless, a disproportionate majority of current intervention research remains narrowly focused on alleviating hyperactivity and inattention, often neglecting the imperative for direct interventions targeting social integration itself. Consequently, to achieve more holistic and sustainable improvements, future paradigms should prioritize the cultivation of pragmatic social skills and the application of robust emotion regulation strategies. Against this theoretical backdrop, the present study systematically explored various exercise modalities and identified marked heterogeneities in their capacity to improve social outcomes.

4.3. Efficacy comparisons across exercise modalities

Among these intervention forms, closed-skill exercise emerged as the most efficacious modality for enhancing social functioning. This superiority may be attributed to the inherent characteristics of closed-skill tasks—specifically their highly structured nature and predictable repetitiveness—which are thought to more potently elicit activation within prefrontal cortex regions. This neural engagement enhances executive control and social coping mechanisms, making it uniquely suited for children and adolescents exhibiting high levels of impulsivity and distractibility (Christiansen et al., 2019). Furthermore, closed-skill exercise have been shown to augment activity in the dorsolateral prefrontal cortex, which facilitates the development of sophisticated social cognition, such as empathy and turn-taking awareness (Benzing et al., 2018). Such structured environments are exceptionally conducive to establishing behavioral regulation habits. In stark contrast, open-skill exercises (e.g., football, basketball), while ostensibly offering more interactive opportunities, impose excessive cognitive demands regarding rapid attention switching and perceptual judgment. For children and adolescents with ADHD, these demands may inadvertently trigger anxiety or impulsive reactions. Under certain conditions, the high-pressure team environment may paradoxically hinder the formation of positive peer bonds, increasing the risk of social exclusion and exacerbating feelings of social frustration, thereby attenuating the overall efficacy of the intervention (Hoza et al., 2015).

4.4. Non-linear dose-response relationships

Regarding the parameters of the intervention, this study identified a non-linear, U-shaped dose-response relationship. This finding implies that moderate-intensity exercise, when delivered at an appropriately calibrated volume, optimizes social skill acquisition, whereas sub-threshold levels fail to provide the requisite cognitive and situational stimulation. Conversely, suprathreshold volumes may induce physiological fatigue, heightened anxiety, or distractibility, which collectively neutralize the intervention's benefits. This complex relationship underscores the necessity of avoiding the "too much of a good thing" pitfall when prescribing exercise regimens for this clinical population.

Specifically, this study proposes 660 MET-min/week as the requisite volume to achieve the MCID. Furthermore, the optimal therapeutic window was identified at approximately 1290 MET-min/week, where social improvements reached their peak. Although this optimal dose is quantitatively lower than the World Health Organization's (WHO) general recommendation of 60 min of daily moderate-to-vigorous activity (Bull et al., 2020), it represents a clinically meaningful milestone for children and adolescents with ADHD. This suggests that even conservative exercise volumes can yield substantial therapeutic dividends, providing a more pragmatic and achievable target for children and adolescents who struggle to meet international physical activity guidelines. Importantly, intervention protocols should be meticulously tailored to individual fitness levels and psychological tolerance, with intensity gradually titrated to maximize efficacy while minimizing the risk of overexertion.

4.5. Influencing factors and research robustness

The moderating influence of individual factors, such as gender and age, warrants specific consideration. Our analysis indicates that female participants exhibited a higher sensitivity to mind-body therapies, whereas males demonstrated superior responsiveness to closed-skill and multi-component formats. This divergence may reflect sex-specific preferences in emotional self-regulation and physical engagement, representing a fertile area for future investigation. Concurrently, the diminished efficacy of certain interventions

observed with increasing age suggests that developmental stages influence intervention acceptance and cognitive adaptability. Therefore, prioritizing early-stage interventions may be instrumental in achieving the most favorable long-term social outcomes.

A discrepancy was observed between the primary network meta-analytic findings and the publication-bias-adjusted estimates derived from the RoBMA framework. Specifically, the primary NMA suggested a favorable overall pattern of exercise effects, whereas the RoBMA-adjusted pooled estimate became statistically uncertain, with the interval including zero. One plausible explanation for this difference lies in the inherently conservative nature of the RoBMA framework. RoBMA applies Bayesian model averaging across competing models that differ with respect to the presence or absence of an overall effect, heterogeneity, and publication bias. Under conditions of substantial between-study heterogeneity and a relatively limited evidence base, the framework may assign greater weight to models closer to the null, thereby yielding a more attenuated pooled estimate.

Although the RoBMA findings suggest that the magnitude of the pooled effect should be interpreted with greater caution, they do not necessarily negate the potential clinical relevance of the present findings. Across different sensitivity models, the direction of the effect remained broadly consistent, indicating a potentially beneficial role of exercise interventions for improving social dysfunction in children and adolescents with ADHD. At the same time, the discrepancy between the primary NMA and the RoBMA-adjusted estimates highlights the need for further high-quality, large-scale randomized controlled trials to provide more precise effect estimates and to reduce the influence of small-study effects and reporting bias.

4.6. Strengths and limitations

This study offers distinctive contributions regarding both methodological rigor and clinical translatability. First, incorporating the MCID as a criterion for assessing social function improvement enhances the clinical interpretability and practical value of the findings. Second, the application of Bayesian network meta-analysis and nonlinear dose-response modeling not only systematically compared the effects of different exercise interventions but also clearly identified effective and optimal dose ranges. Such findings provide robust, quantitative evidence to inform the development of personalized exercise prescriptions, highlighting the study's considerable potential for seamless clinical implementation.

Despite these contributions, several limitations warrant acknowledgment. First, the omission of granular intervention parameters—such as real-time heart rate data, precise intensity metrics, or consistent attendance rates—in some primary studies necessitated the use of proxy-based estimations derived from contemporary literature. Although these procedures were based on established guidelines, they may have introduced additional computational uncertainty. Second, the supplementary publication-bias-adjusted sensitivity analysis suggested a more conservative pooled pattern than conventional assessments alone, indicating that the magnitude of the overall pooled effect should be interpreted prudently. Finally, the evidence base remained demographically imbalanced, with female participants representing only a small proportion of the total sample, and detailed reporting of concurrent medication use was often insufficient, limiting our ability to examine potential treatment interactions.These constraints underscore the need for future research to prioritize more gender-inclusive cohorts and standardized reporting of therapeutic interactions to further refine our understanding of individualized exercise responses.

5. Conclusions

This study demonstrates that exercise interventions can effectively improve social functioning deficits in children and adolescents with ADHD. Specifically, closed-skill exercise yielded the most significant effects, followed by multi-component exercise and mind-body therapy, while open-skill exercise produced relatively limited benefits. Exercise interventions exhibit an optimal dose window, with 660 MET-min/week (equivalent to 16–31 min of moderate-intensity exercise daily) representing the minimum effective dose and 1290 MET-min/week (equivalent to 31–61 min of moderate-intensity exercise daily) constituting the optimal dose. These findings provide quantitative evidence for developing individualized exercise prescriptions in clinical practice, holding significant clinical application value.

CRediT authorship contribution statement

Yiting Wei: Supervision, Resources. **Wanqun Xu:** Supervision, Resources. **Minghui Quan:** Writing – review & editing, Validation, Supervision, Resources, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Huan Huang:** Supervision, Resources, Formal analysis. **Shanru Jin:** Supervision, Resources. **Keyu Pan:** Supervision, Resources, Formal analysis. **Gao Xiang:** Writing – review & editing, Writing – original draft, Visualization, Validation, Investigation, Formal analysis, Data curation, Conceptualization. **Zhang Meng:** Writing – review & editing, Visualization, Validation, Formal analysis. **Changshuang He:** Validation, Supervision, Resources, Project administration, Investigation, Data curation. **Zongyu Fan:** Visualization, Validation, Formal analysis.

Funding

This study was supported by the Key Project of the National Philosophy and Social Science Foundation of China (Grant No. 25&ZD251); the "Dawn" Program of the Shanghai Municipal Education Commission (Grant No. 22SG49); and the Shanghai Oriental Talents Program (Youth, 2023).

Declaration of Competing Interest

The authors do not have any conflicts of interest to disclose.

Acknowledgments

The authors in this review would like to thank the authors of the included articles for sending further information on their studies for the purpose of our review.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.ridd.2026.105276.

Data availability

Data will be made available on request.

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